EE5130: Artificial Intelligence and Machine Learning

Mini Project Guideline

Comparative Study of Supervised Classification and Unsupervised Clustering Techniques Using the MNIST Dataset

Course contribution:	30% of total course marks
Submission:	Technical report and mini exhibition
Deadline:	By the end of the semester

1. Project Overview

The objective of this mini project is to investigate, implement, and evaluate different machine learning techniques using the MNIST handwritten digit dataset. The project consists of two main components:

1. **Supervised learning:** Design and evaluate classifiers capable of recognizing handwritten digits from images.
2. **Unsupervised learning:** Apply different clustering techniques to discover patterns and groupings within the dataset, and interpret the results.

This is an open-ended, experimental project. Students are encouraged to explore a wide range of machine learning algorithms, investigate alternative preprocessing and feature extraction techniques, compare their performance, and draw meaningful conclusions from experimental results.

The emphasis is on practical implementation, systematic experimentation, quantitative evaluation, visualization, and interpretation rather than theoretical descriptions of algorithms.

The project contributes **30% of the total marks for EE5130**. The final submission consists of a technical report and a mini exhibition.

2. Learning Objectives

By completing this project, students should be able to:

- Implement and compare different supervised classification algorithms.
- Apply suitable preprocessing and feature extraction methods to image data.
- Evaluate classifiers using multiple performance metrics.
- Investigate different unsupervised learning and clustering methods.
- Compare clustering results using quantitative measures and visualizations.
- Interpret classification errors, cluster structures, and the limitations of alternative approaches.
- Design and conduct systematic machine learning experiments.

- Communicate experimental findings effectively through a technical report and mini exhibition.

3. Dataset

The project will use the MNIST handwritten digit dataset, which contains grayscale images of handwritten digits from 0 to 9.

Each image has a resolution of 28×28 pixels, giving 784 pixel values when represented as a feature vector.

The standard MNIST dataset contains:

- 60,000 training images.
- 10,000 test images.
- Ten digit classes, labelled 0 to 9.
- Grayscale pixel intensities representing handwritten digits.

Students may use established implementations of MNIST available through TensorFlow/Keras, PyTorch, scikit-learn-compatible datasets, or other suitable sources.

The standard training and test partitions should be preserved when using the conventional MNIST dataset. If computational limitations require a smaller dataset, students may use a documented subset. Final comparisons must clearly state which data and partitions were used.

Digit labels may be used for supervised classification. For unsupervised learning, labels should not be used to train clustering algorithms, although they may subsequently be used to evaluate and interpret the discovered clusters.

4. Part A: Supervised Classification

4.1 Objective

Design classifiers to identify the digit represented by each MNIST image. Investigate different classification techniques and compare their performance using appropriate evaluation metrics.

4.2 Classification Techniques

Students should investigate several classification approaches. The following list provides suggestions rather than mandatory limits.

Classical machine learning methods

- Logistic Regression
- k-Nearest Neighbours (k-NN)
- Support Vector Machines (SVM)
- Decision Trees
- Random Forests
- Naive Bayes
- Linear Discriminant Analysis

- Gradient Boosting or other ensemble classifiers

Neural network methods

- Multilayer Perceptron (MLP)
- Convolutional Neural Network (CNN)
- Other suitable neural network architectures

Students are encouraged to explore additional methods where appropriate. A strong project should include meaningful comparisons between methods with different characteristics rather than simply running many algorithms with default settings.

4.3 Preprocessing and Feature Representation

Investigate suitable preprocessing approaches, such as:

- Normalizing pixel values.
- Flattening images into 784-dimensional feature vectors.
- Retaining the two-dimensional image representation for CNN models.
- Applying dimensionality reduction, such as Principal Component Analysis (PCA).
- Extracting alternative features, such as image gradients or other suitable image descriptors.

Where practical, compare the effects of different preprocessing or feature representation methods on classifier performance.

Any preprocessing operation that learns parameters from data must be fitted using the training data only and then applied to validation and test data. Test data must not be used to select features, tune hyperparameters, or fit preprocessing transformations.

4.4 Model Training and Parameter Selection

Investigate appropriate model parameters and settings. Depending on the method, these may include:

- Number of neighbours in k-NN.
- Kernel and regularization parameters in SVM.
- Tree depth and number of trees in ensemble methods.
- Number of hidden layers and neurons in MLP.
- CNN architecture, batch size, learning rate, and number of epochs.
- Number of principal components used for dimensionality reduction.

Use a validation set or cross-validation where appropriate for model selection. Keep the final test set separate until the selected models are ready for evaluation.

Record the settings used for each experiment so that the results can be reproduced.

4.5 Classification Performance Evaluation

Evaluate each classifier using several appropriate performance measures.

At a minimum, report:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- Confusion matrix.

For multiclass classification, clearly state how precision, recall, and F1-score are averaged. Macro-averaged and weighted-averaged measures are useful for comparison. Micro-averaged measures may also be included.

Other useful measures include:

- Per-digit precision, recall, and F1-score.
- One-vs-rest ROC curves and ROC-AUC, where appropriate.
- Training time and inference time.
- Model size or computational requirements.
- Learning curves and validation performance.
- Performance variation across repeated runs, where practical.

Students should identify which digits are frequently confused and investigate possible reasons for these errors using examples from the dataset.

Include a comparative table of the principal classification results and a discussion identifying the best-performing models under different criteria.

5. Part B: Unsupervised Learning and Clustering

5.1 Objective

Investigate whether unsupervised learning techniques can discover meaningful groupings in the MNIST dataset without using digit labels during clustering.

Compare the structures produced by different clustering algorithms and examine how well the resulting clusters correspond to the handwritten digit categories.

5.2 Suggested Clustering Techniques

Students may investigate several of the following methods:

- k-Means clustering.
- Gaussian Mixture Models (GMM).
- Agglomerative Hierarchical Clustering.

- DBSCAN.
- Spectral Clustering.
- Mini-Batch k-Means.
- Other suitable clustering algorithms.

The number of clusters may be selected based on prior knowledge, exploratory experiments, or model selection criteria. Since MNIST contains ten digit categories, using ten clusters is a useful initial experiment, but students should also investigate alternative settings where appropriate.

Density-based methods may produce a different number of clusters and may classify some samples as noise. These outcomes should be reported rather than artificially forcing every method to produce ten clusters.

5.3 Feature Representation for Clustering

Investigate the effects of different feature representations on clustering performance.

Possible approaches include:

- Original 784-dimensional pixel vectors.
- Normalized pixel values.
- PCA-reduced representations.
- Latent representations extracted using an autoencoder.
- Other suitable feature extraction methods.

Consider the computational cost of clustering high-dimensional image data and investigate whether dimensionality reduction improves the quality or interpretability of the resulting clusters.

5.4 Clustering Evaluation

Use appropriate internal clustering measures to evaluate the structure of the clusters without requiring class labels.

Suggested measures include:

- Silhouette Score.
- Davies-Bouldin Index.
- Calinski-Harabasz Index.
- Within-cluster sum of squares, where applicable.

The interpretation of these measures must take account of the clustering algorithm and data representation. A numerical score alone is not sufficient to establish that a clustering result is meaningful.

Since the true digit labels are available for MNIST, students may also evaluate the correspondence between the discovered clusters and the known classes using external measures, such as:

- Adjusted Rand Index (ARI).
- Normalized Mutual Information (NMI).

- Homogeneity, completeness, and V-measure.

These external measures must be used only after clustering has been performed without digit labels. The labels are for evaluation and interpretation, not for fitting the unsupervised model.

5.5 Visualization and Interpretation

Students should visualize and interpret the discovered cluster structures. Suitable approaches include:

- PCA-based two-dimensional projections.
- t-SNE visualizations.
- UMAP visualizations.
- Cluster-size plots.
- Representative images from each cluster.
- Tables showing the distribution of true digit labels within each cluster.

For each selected clustering method, investigate:

- The number and sizes of the resulting clusters.
- Whether clusters correspond predominantly to individual digit classes.
- Which digits are grouped together and which are separated.
- Whether visually similar handwriting styles appear in the same cluster.
- Whether some digits are distributed across several clusters.
- How dimensionality reduction affects the clustering results.
- The effect of algorithm parameters on the discovered structures.
- Whether the visual appearance of a cluster agrees with the numerical evaluation measures.

Discuss cases in which clustering succeeds or fails to recover the digit categories and consider possible explanations.

Visualizations should be interpreted carefully. A two-dimensional projection may distort distances and cluster structures present in the original feature space.

6. Experimental Design and Comparative Analysis

The project is intended to encourage systematic experimentation. Students should formulate useful comparisons rather than merely execute algorithms.

Possible investigations include:

- Comparing classical machine learning classifiers with neural networks.
- Comparing pixel-based features with reduced-dimensionality features.
- Investigating the effects of different preprocessing techniques.
- Studying the relationship between classification accuracy and computational cost.

- Comparing clustering methods using the same data representation.
- Investigating the influence of the number of clusters and algorithm parameters.
- Comparing clustering results obtained from raw pixels and learned feature representations.
- Examining the relationship between supervised classification performance and unsupervised cluster quality.

Students may investigate additional questions of their own choice.

For fair comparison, clearly document the dataset partition, preprocessing, model configuration, evaluation protocol, and relevant parameter settings. Where stochastic algorithms are used, repeated runs may be appropriate.

The number of experiments is less important than the quality of the experimental design, the reliability of the evidence, and the insight gained from the results.

7. Technical Report

Submit a technical report summarizing the experiments, results, and conclusions. The report should be concise, evidence-based, and organized so that another student can understand and reproduce the main experiments.

A theoretical review of machine learning algorithms is not required. Provide only brief descriptions of the methods and implementation choices necessary to understand the experiments.

The report should contain the following sections.

7.1 Introduction and Objectives

Briefly state the aims of the project and the questions investigated.

7.2 Experimental Setup

Describe the dataset, software libraries, computing environment, data partitions, preprocessing, feature representations, and experimental procedures.

7.3 Classification Experiments

Present the classifiers investigated, relevant parameter settings, evaluation metrics, and comparative results.

Include confusion matrices and other visualizations where they provide useful information.

7.4 Clustering Experiments

Describe the clustering methods, feature representations, parameter settings, and evaluation procedures.

Present internal and external clustering measures, cluster visualizations, and representative examples where appropriate.

7.5 Comparative Results and Discussion

Use tables and figures to compare the methods investigated.

Discuss the main findings, unexpected results, important errors, computational costs, and trade-offs between alternative methods.

Identify the most effective methods for the particular objectives studied and explain the evidence supporting these conclusions.

7.6 Conclusions and Future Work

Summarize the main findings and identify promising improvements or additional experiments.

7.7 References

Cite datasets, software libraries, external resources, and publications used in the project. Follow a consistent referencing style.

7.8 Reproducibility Materials

Provide the source code, configuration details, and instructions needed to reproduce the principal results. These may be submitted as an appendix, a code repository, or supplementary electronic material, as specified by the course coordinator.

All figures and tables should have appropriate captions. Report actual experimental results rather than illustrative or invented values.

8. Mini Exhibition and Final Presentation

The final presentation will take the form of a mini exhibition. Students should demonstrate their work and communicate the main findings in a clear, engaging, and technically sound manner.

The exhibition should ideally include:

- A brief statement of the problem and project objectives.
- An overview of the experimental workflow.
- Demonstrations of selected classifiers and their predictions.
- Comparative classification performance tables or visualizations.
- Confusion matrices and examples of classification errors.
- Demonstrations of selected clustering methods.
- Visualizations of the discovered clusters.
- Examples of correctly grouped, incorrectly grouped, and ambiguous digits.
- Key findings, limitations, and conclusions.

Where practical, students should prepare a working demonstration that allows visitors to supply an image or select a sample and observe the predicted digit or its assigned cluster.

Posters, displays, notebooks, dashboards, and interactive demonstrations may be used. The exhibition should prioritize the clarity and interpretation of results over decorative presentation.

Students should be prepared to explain their implementation choices, demonstrate the operation of their systems, interpret evaluation measures, and justify the conclusions drawn from their experiments.

9. Assessment Scheme

The mini project contributes **30% of the total course marks**. The following rubric allocates 100 marks within the mini project.

Assessment component	**Marks**
Classification implementation and experimental breadth	20
Classification evaluation and comparative analysis	20
Clustering implementation and experimental breadth	15
Clustering evaluation, visualization, and interpretation	20
Technical report quality and reproducibility	15
Mini exhibition, demonstration, and technical communication	10
Total	**100**

The corresponding course contribution is calculated as:

$$\text{Course contribution} = \frac{\text{Mini project marks}}{100} \times 30$$

The assessment will consider the quality of implementation, suitability of evaluation methods, fairness of comparisons, depth of interpretation, and clarity of communication.

Credit will be given for thoughtful experimentation and evidence-based conclusions, including well-explained negative results. Simply running many algorithms without meaningful evaluation or interpretation will not be sufficient for a high mark.

10. Submission Requirements

Each project submission should include:

1. A technical report containing the experimental results and conclusions.
2. Source code and instructions for reproducing the principal experiments.
3. Relevant tables, figures, confusion matrices, and clustering visualizations.
4. The materials required for the mini exhibition.
5. A record of the main experimental settings and any important limitations.

Submit the report and associated electronic materials by the end of the semester, according to the instructions provided by the course coordinator.

The report file format, submission platform, presentation duration, and any additional requirements will be communicated separately.

11. Good Practices and Academic Integrity

Students are expected to:

- Use the official dataset partitions appropriately and avoid data leakage.
- Report actual results and identify the evaluation protocol used.
- Distinguish between training, validation, and test performance.
- Explain limitations caused by using subsets of the dataset.
- Cite external code, libraries, datasets, and resources.
- Clearly identify externally sourced code and explain the modifications made.
- Ensure that the submitted implementation and results are understood by the students presenting them.
- Follow the university's academic integrity requirements.

AI-assisted programming tools may support development and experimentation, subject to university policy. Students remain responsible for verifying their code, validating their results, acknowledging assistance where required, and understanding the work they submit.

12. Final Remarks

This project is an opportunity to explore machine learning beyond a prescribed set of exercises. Students are encouraged to be curious, investigate unexpected results, test alternative approaches, and develop their own research questions.

The most successful projects will demonstrate not only that machine learning algorithms can be implemented, but also that their performance can be evaluated critically, their limitations understood, and their results interpreted meaningfully.

The central objective is to learn through experimentation and to communicate what the experiments reveal.